Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Cloud and Tools: Model serving, Recommendation systems, Hypothesis testing, Regression analysis, TensorFlow, A/B testing, Feature engineering, Docker, Git, PySpark, Vector embeddings, Cross-validation, Hyperparameter tuning
Machine Learning and AI: Computer vision, PyTorch, Scikit-learn, Pandas, NumPy, LSTM, Flask
Languages: Python, SQL
Data Engineering: Apache Kafka, Apache Flink, Apache Spark, ETL/ELT

Projects

Real-Time Rider-Driver Matching Model *Python, PyTorch, Apache Kafka, Apache Flink*

- Built a deep learning matching model in PyTorch that scores rider-driver pairs from session and geospatial features, designed for low-latency inference on live request streams
- Implemented a streaming feature pipeline with Apache Kafka and Apache Flink to compute up-to-date driver availability and demand signals consumed at scoring time
- Structured offline evaluation with hypothesis testing and regression-based error analysis to compare candidate model variants against marketplace KPIs

MiGa - Multi-Chicken Gait Assessment (AICV Lab) *YOLO11n-Pose, OC-SORT, LSTM, PyTorch*

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking, and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a detection split of 4,569 frames with 11,422 bounding boxes and a regression split of 55 expert-annotated sequences
- Implemented spatial normalization into local coordinate spaces to eliminate translational variance, achieving keypoint mAP 91.83% at 1,250 FPS, 0.93 MOTA, and 0.91 IDF1

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics
- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation